

MODULE - I

(Short Answer Type Questions)

Each of 5 marks

1. Explain database schema with example.
2. Define (i) Database (ii) DBMS
3. Discuss five main advantages of database management system over file system.
4. Explain the role of Database Administrator.
5. Explain different types of database end users.
6. Explain database instance with example.
7. Explain logical data independence and physical data independence with suitable diagram.
8. What is the purpose of data base system? Describe the three levels of data abstraction.
9. What are the different types of schema?
10. Explain the database design process in detail.
11. Explain the difference between meta data and data dictionary with example. What are the different data models?
12. Discuss about Database users and Administrators.
13. Explain about Database languages with examples.

(Long Answer Type Questions)

Each of 10 marks

1. Discuss Three Tier Database Architecture with diagram.
2. Define the terms 1) physical schema 2) logical schema 3) Conceptual Schema with suitable diagram.
3. Draw the Architecture of Database.
4. What are the two types of participation constraint? What are different access types in DML?
5. Differentiate File Processing System with Database Management System.

MODULE - II

(Short Answer Type Questions)

Each of 5 marks

1. Explain Derived attribute and Multivalued attribute with example and diagram.
2. What is Key Attribute? Explain with example.

3. What is foreign key ? Explain with example.
4. Explain Hierarchical Data Model with example.
5. Define Entity, Attributes, Entity set, relationship with appropriate notations in ERD. Draw the notation for multivalued attributes. Give one example.
6. Draw ER diagram for Ternary Relationship set with suitable example.
7. What does the cardinality ratio specify? List any eight applications of DBMS.
8. Define Relational Database. Explain data representation in RDBMS.
9. What is referential integrity constraints? How it is related to foreign key. Explain with example.

(Long Answer Type Questions)

Each of 10 marks

1. Define Weak Entity set, candidate key, primary key and foreign key.
2. Explain Generalization and Specialization with example.
3. What is the difference between Hierarchical and Network Model? Explain with example.
4. Explain about integrity constraints over relations.
5. Draw the ER diagram for a company needs to store information about employees(Identified by ssn, with salary and phone as attributes), departments(Identified by dno, with dname and budget as attributes), and children of employees(With name and age as attributes). Employees work in departments, each department is managed by an employee, a child must be identified uniquely by name when the parent(Who is an employee; assume that only one parent works for the company) is known. We are not interested in information about a child once the parent leaves the company.
6. Create ER diagram for each of the followings:

Each company operates four departments and each department belongs to one company. Each department in part (a) employes one or more employees and each employee works for one department. Each of the employees in part(b) may or may not have one or more dependents.

MODULE - III

(Short Answer Type Questions)

Each of 5 marks

1. Explain Data Definition Language (DDL) with suitable commands.
2. Write suitable SQL query to display minimum, maximum, average, and total salaries earned by employees.

3. Explain Data Manipulation Language (DML) with suitable commands.
4. Explain projection operation in relational algebra with suitable example.
5. State the difference between cartesian product and natural join in context of relational algebra.
6. Give an example of how to use the rename operation in relational algebra.
7. Explain the following
 - i. Data Definition Language.
 - ii. Data control Language.
 - iii. Views
8. What is a view? How can it be created? Explain with an example.
9. What is a join operator? Explain about conditional join and natural join with syntax and example.
10. How can we compare using null values? Explain about logical connectives with examples.

(Long Answer Type Questions)

Each of 10 marks

1. Discuss different types of outer join with example.
2. Discuss different types of integrity constraint with example.
3. i) Write suitable relational algebra operations to display columns Regno, Section from the table Students where branch is EE.
 ii) Write suitable relational algebra operation to select tuples from the table Book where the topic is "Database" and author is "Ram".
4. What are aggregate functions? Explain five built-in aggregate functions.
5. Explain the structure of basic form of an SQL query. What is cursor in database? How can you create an user defined cursor to store all manager details in SQL?
6. Consider the given relations:
 STUDENT(SID, Name, Dept, Age)
 COURSE(CID, CName, Dept)
 ENROLL(SID, CID, Marks)
 Write Relational Algebra expression for the following queries:
 - Find names of all students.
 - Find names of students whose age is greater than 20.
 - Find students enrolled in DBMS.
 - Find students who are not enrolled in any course.
 - Find students who are enrolled in all courses.
7. What are the differences between natural join and cross product of two relations? Explain with example. Discuss each of followings- (a) Natural Join b) equi-join c) self join d) inner-join e) outer join

MODULE - IV

(Short Answer Type Questions)

Each of 5 marks

1. Explain Functional dependency with suitable example.
2. Explain transitive dependency with suitable example.
3. Define Third Normal Form (3NF) with example.
4. What is redundancy? Explain with suitable example. What are the anomalies created by data redundancy? Explain.
5. Explain the classification of functional dependency.
6. Consider the relation schema $R(A,B,C,D,E,F)$ and the functional dependencies $A \rightarrow B, C \rightarrow DF, AC \rightarrow E, D \rightarrow F$. Find the candidate keys of this relation R .
7. Define and explain 4NF with suitable example.
8. Explain join dependencies with fifth normal form with an example.
9. Consider the relation $R(v,w,x,y,z)$ with functional dependencies $\{z \rightarrow y, y \rightarrow z, x \rightarrow y, x \rightarrow v, vw \rightarrow x\}$. Find the closure for $\{xy\}$ given functional dependencies.
10. Given below are two sets of FD's for a relation $R(A,B,C,D,E)$. Are they equivalent?
 $F = \{A \rightarrow C, AC \rightarrow D, E \rightarrow AD, E \rightarrow H\}$ and $G = \{A \rightarrow CD, E \rightarrow AH\}$
11. "All BCNF relations are in 3 NF but not vice versa". Prove the statement with example.
12. Explain Multivalued dependency with suitable example.

(Long Answer Type Questions)

Each of 10 marks

1. Consider a schema $R(A,B,C,D)$ and functional dependencies $A \rightarrow B$ and $C \rightarrow D$. Check that the decomposition of R into $R_1(AB)$ and $R_2(CD)$ gives lossless join and dependency preserving decomposition or not.
2. Consider a relation $R(A,B,C,D,E,F,G)$ with the functional dependencies
 $AB \rightarrow CD$
 $AF \rightarrow D$
 $DE \rightarrow F$
 $C \rightarrow G$
 $F \rightarrow E$
 $G \rightarrow A$
 Find closure of attribute sets $\{BG\}$ and $\{AF\}$. Illustrate fully functional dependency with example.
3. Consider the following relation: Shipping (Shipname, Shiptype, Voyageid, Cargo, Port, Date) (note: Date the date the ship arrives in the given port) with functional dependencies-
 $\{ShipName \rightarrow Shiptype, Voyageid \rightarrow Shipname, Cargo, Shipname, Date \rightarrow Voyageid, Port\}$
 a) Identify the candidate keys.
 b) Normalize the relation until it is completely normalized.
4. Define BCNF. How does it differ from 3 NF ? Why is it considered as stronger than 3 NF ?
5. Discuss Armstrong's Axioms.
6. Consider the following relation schema
 $Works(Pname, Cname, salary)$
 $Lives(Pname, Street, City)$
 $Located_in(Cname, city)$
 $Manager(Pname, Mgrname)$

Write the SQL queries for the following.

- i) Find the names of all persons who live in the city Bangalore.
- ii) Retrieve the names of all person of "Infosys" whose salary is between Rs .50000
- iii) Find the names of all persons who lives and work in the same city
- iv) List the names of the people who work for “Tech M” along with the cities they live in.
- v) Find the average salary of “Infosys” persons

MODULE – V

(Short Answer Type Questions)

Each of 5 marks

1. Explain the pros and cons of Sequential File Organization.
2. Write the differences between B-Tree and B+ Tree.
3. Explain the concept of Hash File Organization with suitable diagram.
4. Define the terms in the context of file and storage management. Seek time, access time, rotational latency.
5. Compare Primary indexing and clustering indexing in file management techniques.
6. Disadvantages of B TREE over B+ TREE.
7. Explain how secondary indexing can help in join operation.

(Long Answer Type Questions)

Each of 10 marks

1. Explain the concept of Sequential File Organization with suitable diagram.
2. Consider insertion sequence- 2,3,5,11,17,19,23,29,31. Construct B+ Tree for the case where the number of pointers that will fit in one node is 4.
3. Given a block can hold either 3 records or 10 key pointers. A database contains n records, then how many blocks do we need to hold the data file and the dense index.
4. The order of a leaf node in a B+ tree is the maximum number of (value, data record pointer) pairs it can hold. Given that the block size is 1K bytes, data record pointer is 7 bytes long, the value field is 9 bytes long and a block pointer is 6 bytes long, what is the order of the leaf node?

MODULE - VI

(Short Answer Type Questions)

Each of 5 marks

1. Explain serial schedule with example.
2. Explain non-serial schedule with example.
3. Explain Transaction Control Language (TCL) with suitable commands.
4. Define Lock. Discuss Two-Phase Locking protocol with suitable diagram.
5. When a schedule is called non-recoverable schedule? Explain.
6. Differentiate between Serial Schedule and serializable schedule. Explain different types of locks.

(Long Answer Type Questions)

Each of 10 marks

1. Explain the ACID properties of transaction with example.
2. Discuss the states of transaction with suitable diagram.
3. Explain Lock-based protocols in details.
4. Draw the precedence graph for the schedule w1 (A) w2 (A) r3 (B) w3 (B) r1 (B):
S: {r1(x); r2(y); r3(y); w2(y); w1(x); w3(x); r2(x); w2(x)}
Test whether the schedule is conflict serializable or not? If Yes, then what is the correct order of serialization.